

Kumbara by Sembol

• TESTNET

A self-custodial dollar piggy bank for Türkiye, opened with Face ID.

Live (Stellar TESTNET)

kumbara.sembol.xyz

Repository

github.com/keyboard01/kumbara

Deck date

6 September 2026

Lira savings lose value; the dollar route has a wall in front of it.

Inflation makes lira saving a losing move

Consumer prices in Türkiye rose **31.51% year on year** in August 2026.¹ Dollar-denominated saving is how ordinary households defend themselves.

The crypto route exists but sheds users at every step

USDC on Stellar settles in seconds for fractions of a cent. Getting there means an extension or app install, twelve words to keep safe, XLM for gas bought somewhere else, and a phone you must never lose.

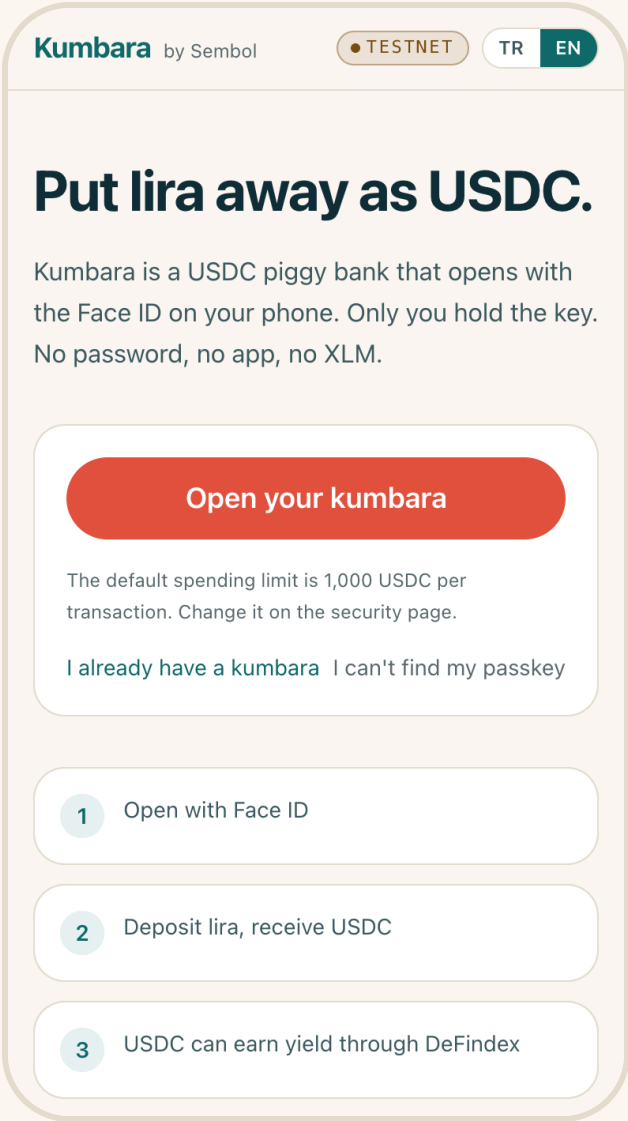
The people who need hard-currency savings most are served least

A bank-transfer habit and a phone with Face ID are what people have. That is the whole interface Kumbara asks for: no seed phrase, no XLM, no install.

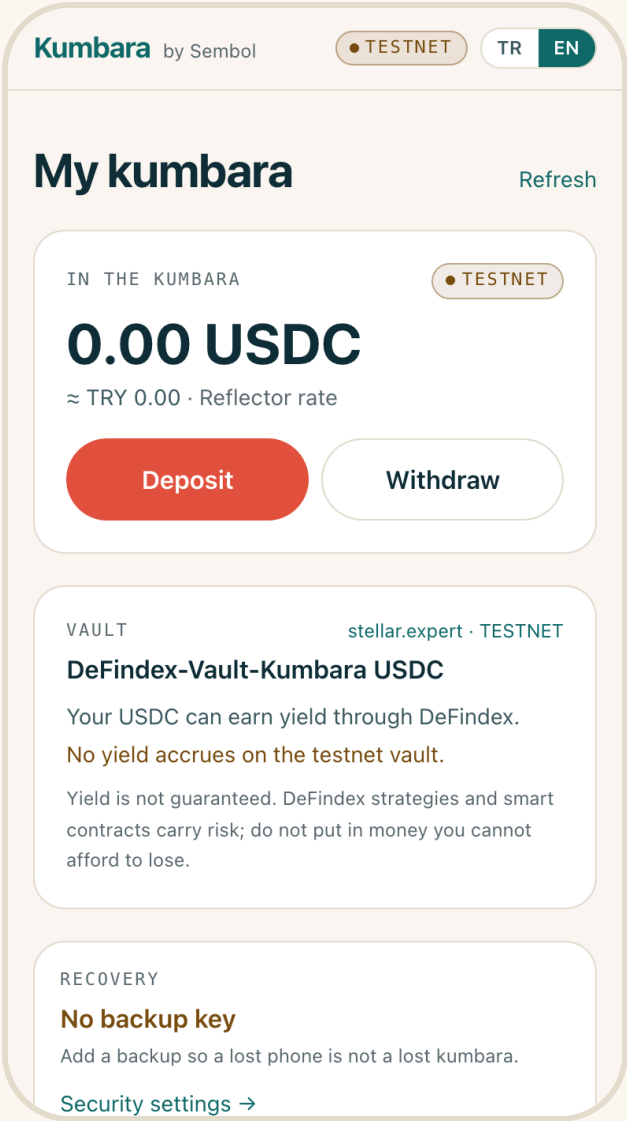
Türkiye is market one. The product is built for a class of markets: high-inflation economies with a bank-transfer habit and an existing Stellar anchor. The anchor is configuration, not code, so the same app serves the next such market.

¹ TÜİK (Turkish Statistical Institute), *Tüketici Fiyat Endeksi, Ağustos 2026*, news bulletin of 3 September 2026: CPI +31.51% against August 2025 (monthly +1.84%). Listed on tuik.gov.tr ("%31,51 (yıl)") with the bulletin at veriportal.tuik.gov.tr/tr/press/58290; verified on tuik.gov.tr on 6 September 2026.

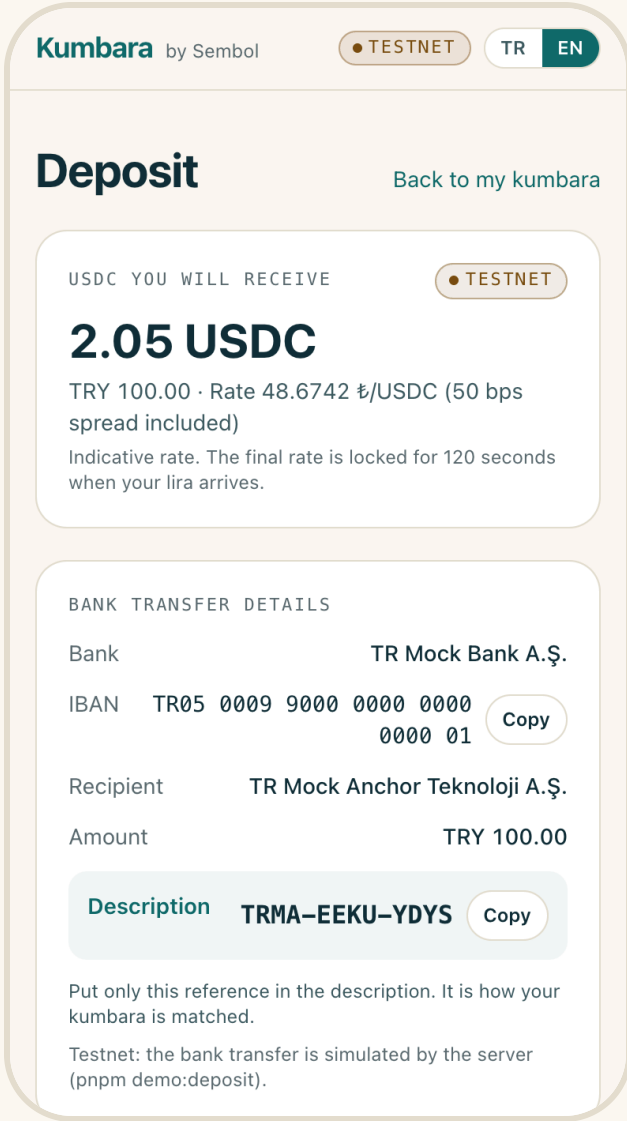
Four screens. Tap to account: ~10 s. No seed phrase, no XLM, no install.



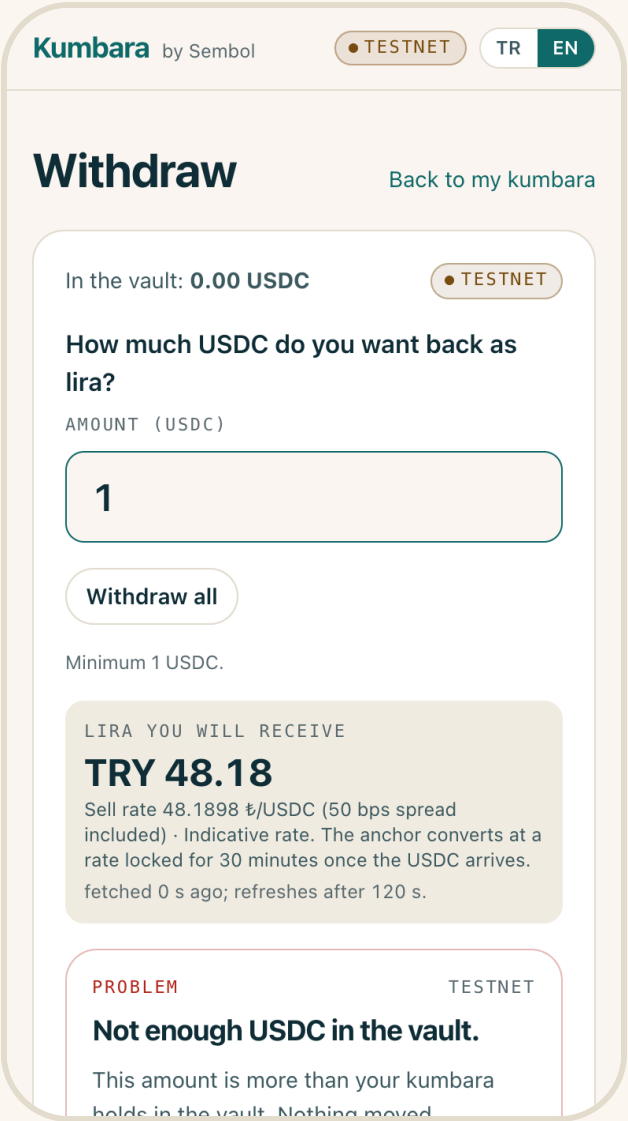
Onboard. One button; Face ID creates the key and the relay deploys an OpenZeppelin smart account. Measured 8–16 s tap to Savings in browser runs.



Savings. USDC in the DeFindex vault with its lira equivalent (Reflector rate), spending limit, recovery. Every figure carries the TESTNET badge.



Deposit. IBAN, recipient and a reference code, exactly like an online bank transfer; the anchor converts lira to USDC.



Withdraw. A sell quote in lira; two Face ID approvals; the anchor pays the visitor's own IBAN.

Scan, tap once, and you have a kumbara before this slide is over.



kumbara.sembol.xyz ● TESTNET

1. Scan the code (it opens the app with this booth's reference).
2. Tap **Kumbaranı aç** and approve with Face ID: an OpenZeppelin smart account is deployed for you; the relay pays the fees.
3. Tap **Yükle**, enter 100, and the presenter plays the bank: about a minute later the USDC is in the DeFindex vault.
4. Tap **Çek** to send it back as lira; the payout reference appears in under a minute.

iPhone: Safari, not private browsing. Android: Chrome with a screen lock. Nothing to install, nothing to write down, no XLM. The counter on the booth screen and kumbara.sembol.xyz/stats move as you go.

HOW IT WORKS

Seven steps, one flow: anchor in front, DeFindex behind, a fee-sponsoring relay underneath.

1

The visitor opens an OpenZeppelin smart account with Face ID; the relay pays the fees.

2

A spending limit (1,000 USDC per transaction) is written on-chain in the background.

3

The anchor issues an IBAN and a reference; the visitor sends lira (simulated by the server on testnet).

4

The anchor converts the lira to USDC and pays an ownerless, pre-authorized bridge account.

5

The bridge account forwards the USDC to the kumbara and closes itself; nobody holds its key.

6

One Face ID puts the USDC into the DeFindex vault (no yield on testnet).

7

A withdrawal runs the path backwards: vault → bridge account → anchor → lira to the IBAN.

The full sequence diagram, every message from the passkey to the payout, is in the appendix (last slide).

Anchors pay classic accounts. Smart accounts are contracts. Ramp payouts could not reach them.

The finding

The anchor's on-ramp rejects a contract (C...) destination and its watcher matches classic payments only: a payout cannot land in a smart account, and a smart account's Soroban transfer is never matched on the way back.

The solution: ownerless, pre-authorized bridge accounts

A throwaway classic account whose only valid transactions are signed before it exists: the **forward** at sequence +1 (USDC to the user's smart account, the exact quoted amount) and the **cleanup** at sequence +2 (trustline dropped, reserves back to the sponsor). The master key is wiped in-process: **no key can sign anything else**. In reverse, the same primitive carries withdrawals to the anchor's treasury with the memo id.

10 / 10

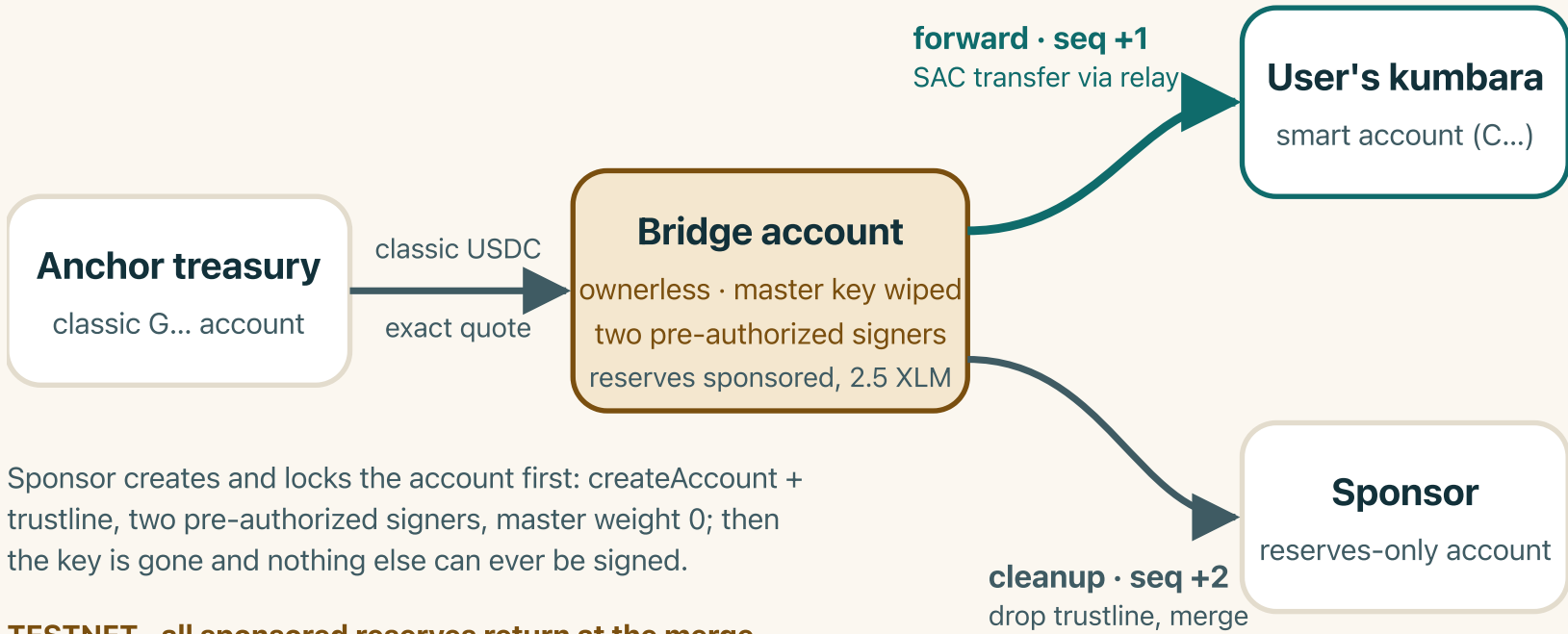
exact amount matches in ten scripted deposits

900

stroops net per deposit for the sponsor; reserves return at cleanup

4

signatures the bridge key ever produces, asserted by a unit test



Sponsor creates and locks the account first: createAccount + trustline, two pre-authorized signers, master weight 0; then the key is gone and nothing else can ever be signed.

TESTNET · all sponsored reserves return at the merge

EVIDENCE (STELLAR TESTNET, STELLAR.EXPERT/EXPLORER/TESTNET/TX/...)	TRANSACTION
Anchor pays the bridge account (classic payment)	778215b3...e3c3a8f1
Pre-authorized forward: USDC lands in the smart account	f580e030...f22c9eab
Pre-authorized cleanup: trustline removed, account merged	6223c61c...7416291d

Reported to the anchor team with the two fixes that would retire the bridge: SAC delivery to contract wallets and Soroban-aware matching on the way back. Full hashes in docs/anchor-notes.md.

Listed building blocks, wired end to end. Kumbara writes no wallet, vault or strategy code.

BUILDING BLOCK	WHAT IT DOES IN THIS APP	ADDRESS (TESTNET, STELLAR.EXPERT)
OpenZeppelin smart accounts + spending-limit policy	Every kumbara is an audited smart account whose only signer is the visitor's passkey; a USDC-scoped rule caps 1,000 USDC per transaction on-chain. Via @sembol/passkey-react and stellar/smart-account-kit.	WASM 1b5f4534...97785a · policy CABXBYJN...2TIP5G
DeFindex vault + hodl strategy	Deposits go through the vault's real invest path into its strategy and withdrawals unwind from it. The strategy is DeFindex's own hodl crate, built unmodified; no yield on testnet, none promised.	vault CAT76PQM...MKSKL · strategy CDFLOE4H...JLDMQN
TR Mock Anchor · SEP-1/6/38 + Partner API	The regulated lira side: IBAN and reference, quotes with a 120 s lock, on-ramps to the bridge account, off-ramps by memo, simulated FAST payouts. Issuer and endpoints read from its stellar.toml.	USDC SAC CBIELTK6...XQDAMA
Fee-sponsoring relay (OpenZeppelin Channels today)	Signed authorization entries go to our server route, which adds the project key and forwards to the relay; the visitor never sees XLM. Per-IP and per-booth caps, sponsor threshold.	channels.openzeppelin.com/testnet
Reflector	The USD/TRY foreign-exchange oracle behind the lira equivalent on the Savings and stats pages (read by simulation, anchor rate as fallback).	CBKGPWGK...4DXMCJZC (mainnet oracle, read only)

Upstream: "Fix wallet creation crash on default Next.js apps" was merged into [stellar/smart-account-kit](#) on 19 August 2026 and shipped in 0.6.2 · [github.com/stellar/smart-account-kit/pull/4](#)

Open-source plan: the repository is MIT-licensed. Kumbara deploys no proprietary contracts: the smart account, WebAuthn verifier and spending-limit policy are OpenZeppelin's audited contracts, the vault and strategy are DeFindex's; the bridge-account logic (`lib/landing`) is MIT in the repo with its tests, and the repository goes public at award time.

Everything on this slide is public, linkable and re-runnable.

Kumbara by Sembol

TESTNET

TR

EN

Kumbara live

live updated 1:17:47 AM

Public, read-only. Testnet: no real lira moves.

Since event start

All time

1

kumbaras opened

TESTNET

1

deposits completed

TESTNET

0

withdrawals completed

TESTNET

TRY 100.00

TRY in

TESTNET

TRY 0.00

TRY out

TESTNET

26.79

USDC

USDC in the vault (live)

TESTNET

kumbara.sembol.xyz/stats: headline numbers with a TESTNET badge, live feed, onboarding curve, timings. The *since event start* / *all time* toggle counts from 19 September 08:00 and shows all-time until then; seed and E2E accounts are excluded by default.

6

real-browser round trips in CI: onboard, deposit, withdraw, booth, stats, failure screens · every push and every 6 h against production

25

failure states, each a TR/EN screen with one action; every flow resumes from stored state

94 · 100 ·
100 · 100

Lighthouse mobile on /stats: performance, accessibility, best practices, SEO

E2E · passing · latest production run 6
Sep 2026

GitHub Actions, workflow e2e.yml (repository badge available once the repo is public)

ON-CHAIN EVIDENCE (STELLAR TESTNET)

TRANSACTION

Vault deployed through the DeFindex factory with the hodl strategy attached

dbf0d5b7...6faa8f96

Deposit with invest=true moved USDC into the strategy (invest)

b615e7dc...76b7f981

Withdrawal unwound the shares from the strategy (divest)

466f48d2...07a9f5ca

Recorded fallback demo: docs/demo/round-trip.mp4, the full round trip on kumbara.sembol.xyz, 2:39 with a caption per step.

Non-custodial software in front of a licensed anchor. Built for any market that has one; Türkiye is market one.

- **Kumbara never holds lira, USDC or keys.** Every transfer is authorized by the visitor's own passkey on their own smart account; the bridge accounts have no owner.
- **The licensed anchor is the regulated party:** it holds the fiat, performs KYC and pays out only to the customer's own named bank account.
- **No payments feature:** no send, no P2P, no merchant flow. Deposit into a vault and withdraw back to your own account, nothing else.
- **Local AML rules shape the UX** wherever it launches: named accounts in and out, an on-chain per-transaction cap, anchor-side limits, no cash or third-party paths.
- **The regulatory gate is the anchor's own position:** the selected anchor's written compliance position on a non-custodial front end, obtained during the Tranche 1 go/no-go. No paid counsel; no entity is needed to receive the award.

ANY LICENSED STELLAR ANCHOR · THE ANCHOR IS A CONFIG CHOICE

Issuer and endpoints come from the anchor's stellar.toml, deposits and withdrawals run on SEP rails, and the bridge accounts pay any classic account, so they are anchor-agnostic. The vault and wallet layer never change. Per market: copy, currency display and the rate oracle.

Market class: high-inflation economies with a bank-transfer habit and an existing Stellar anchor. **Türkiye first, by fit:** 31.5% annual inflation (TÜİK, August 2026), a transfer habit everyone already has, and the TRY sandbox rails this deck demonstrates.

GO / NO-GO IN TRANCHE 1 · FALLBACK BUILT IN

Tranche 1 ranks the on/off-ramps on the SCF Integration List by market: licensed status, bank-transfer rails to the customer's own account, a written non-custodial position, and whether a counterparty entity is required (a shortlist criterion, not a cost). Türkiye stays first only with a licensed TRY anchor that passes; otherwise the pilot launches in the top-ranked market and the app does not change.

The same shortlist is the market-two shortlist. A second market is the natural follow-on award once the metric is hit; it is not promised inside this one.

One builder, already shipping in the ecosystem.

Ahmed

Founder · builds and pitches

Instaward grantee (Stellar Türkiye). Author of [@sembol/passkey-react](#), the passkey wallet layer Kumbara runs on, and contributor to [stellar/smart-account-kit](#). Designed and built Kumbara end to end: the app, the bridge accounts, the CI round trips and the evidence in this deck.

Site	ahmed.bet
LinkedIn	linkedin.com/in/ahmed-zm
GitHub	github.com/keyboard01

SHIPPED UPSTREAM

stellar/smart-account-kit #4: "Fix wallet creation crash on default Next.js apps", merged 19 August 2026, in release 0.6.2.

PUBLISHED

@sembol/passkey-react on npm: passkey creation and signing, recovery, spending-limit policy UI. Every screen in this deck runs on it.

FUNDED

Instawards grantee. Kumbara is the consumer proof for that work, built for this submission.

Groundwork, testnet-complete and secured, then mainnet in the selected pilot market. The ask, one engineer at 60% plus contractor lines: \$54,810.

DELIVERABLE	HOW REVIEWERS VERIFY	WHO · HOURS	USD
TRANCHE 1 · GROUNDWORK · 60 DAYS			
Anchor shortlist and go/no-go: Integration List on/off-ramps ranked by market on licence, own-account bank rails, written non-custodial position, entity requirement; pilot anchor and market selected	ranked shortlist, decision memo, the anchor's written position (repo)	me · 206 h	\$14,420
Audit Bank application with scope: policy install, bridge accounts, server path	submission confirmation; scope document	me · 12 h	\$840
Selected anchor wired in via its sandbox or API in place of the mock anchor (SEP-6/24, SEP-38 where offered); bridge accounts unchanged	sandbox round trip green in CI	me · 70 h	\$4,900
Relay budgets and allowlists (per-project caps, contract allowlist, refusal alerts); hosted and self-host deployment docs with scripted setup	config and docs in repo; a refused call's tx; a fresh deploy	me · 74 h	\$5,180
TRANCHE 2 · TESTNET-COMPLETE AND SECURED · 60 DAYS			
Threat model in SDF's STRIDE template format (assets, threat table, mitigations)	docs/security/threat-model.md	me · 36 h	\$2,520
Monitoring plan in the builders template; core alerts for the vault, policy, sponsor and bridge accounts implemented by me (the gate)	monitoring-plan.md; a live alert on testnet	me · 50 h	\$3,500
↳ Monitoring, full coverage: dashboards, alert routing, runbook. Parallel; without a hire the tranche closes on the core alerts, unbilled	dashboard URL; a routed alert	contractor · 60 h	\$3,300
Audit Bank audit of the policy install, bridge accounts and server path: our scoping and support; findings remediated with regression tests (fee: Audit Bank)	audit report; fix commits; auditor re-check	me · 80 h	\$5,600
Mainnet path behind a capped, presenter-only flag: real USDC, enforced budgets	mainnet tx hashes from the capped path	me · 40 h	\$2,800
TRANCHE 3 · MAINNET LAUNCH AND PUBLIC PILOT, SELECTED PILOT MARKET · 60 DAYS			
Mainnet launch in the selected pilot market by week 3: contracts registered, sponsor account and relay budgets live, the anchor's production rails	mainnet contract IDs on stellar.expert	me · 70 h	\$4,900
Public pilot cohort recruited and run by me; validation-focused user testing (within the 1/3 exception)	/stats on mainnet; tx hashes; cohort report	me · 50 h	\$3,500
↳ User-testing ops: structured sessions, interviews, write-ups. Parallel; without a hire the report comes from /stats data alone, unbilled	session log; cohort report	contractor · 50 h	\$2,750
Metric window and pilot support: 30-day NAV measurement, public dashboard, footprint registered; fixes during the window	NAV dashboard URL; registered IDs; issue log	me · 86 h	\$6,020
↳ Minor, contractor only: native mobile (Expo) SDK for the passkey layer, with docs. Not a gate; billed only if delivered, dropped if no hire lands	npm package and docs URL	contractor · 100 h	\$5,500
Total ask · Integration Track Medium band (\$50k–100k) at its floor, under \$60k on purpose: sized to the scope, not the cap · without the optional Expo line: \$49,310		618 h + 210 h hired	\$54,810

FINAL-TRANCHE METRIC COMMITMENT

Metric type

DeFindex vault NAV: USDC in Kumbara's mainnet vault position, from local-currency deposits in the pilot market

Threshold

average daily NAV of **5,000 USDC** = 200 pilot users × 50 USDC average balance × 50% still holding through the window

Window

the 30 days after the pilot launch, averaged over the window, not a snapshot

Footprint, registered at award time

vault, strategy and policy contracts; sponsor account (bridge accounts sponsored per CAP-33); app-created smart accounts

The panel ratifies or adjusts the threshold during panel review. Seed and E2E accounts are tagged out; wash or gamed activity is excluded.

CANDIDATE ANCHORS · SCF INTEGRATION LIST · UNRANKED UNTIL TRANCHE 1

Koywe · Argentina, Brazil, Chile, Colombia, Mexico, Peru, Bolivia, United States · sandbox; API in 1–2 weeks

alfredpay · stablecoin-to-banking rails across Latin America · **Abroad** · local instant payment rails · 1–2 weeks

BlindPay · fiat and stablecoin transfers worldwide · 1–2 weeks · **Mercuryo** · fiat-to-crypto on/off-ramp · 1–2 weeks

Bridge · multi-currency payments settled on Stellar · 1–5 days

Türkiye: the TRY rails in this deck are a sandbox and no TRY anchor is on the list today; Türkiye stays market one only if a licensed TRY anchor passes the same criteria. MoneyGram (cash) and Etherfuse (bonds) are listed but outside the bank-transfer shape.

BUDGET HYGIENE

Only development, testing, monitoring, deployment and validation work is in the ask; nothing is self-funded. No legal fees, no entity setup (none is needed to receive the award), no audit fee (Audit Bank), no marketing beyond the user-testing exception. Contractor lines are parallel capacity, never dependencies: each has a no-hire outcome that still closes the tranche. Nothing waits on a third party signing.

Every message in the round trip, from the passkey to the payout.

